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Force transmission mechanism for surgical instrument, and related devices, systems, and methods

Abstract

An instrument includes a movable end effector component coupled to a shaft; a force transmission mechanism; and an actuation mechanism coupled to the force transmission mechanism and the end effector, the actuation mechanism comprising a first actuation member segment and a second actuation member segment. The force transmission mechanism comprises: a first pulley coupled to a rotatable drive input and having a first axis of rotation, and a second pulley having a second axis of rotation non-parallel to the first axis of rotation of the first pulley. The first actuation member segment is routed over the first pulley. In response to rotation of the rotatable drive input, at least a portion of the first actuation member segment is wound onto the first pulley and unwound from the second pulley, and the second actuation member segment is translated along the shaft to actuate the movable end effector component.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/708,544, filed Mar. 30, 2022, which is a continuation application of U.S. application Ser. No. 16/083,150, filed Sep. 7, 2018 (now U.S. Pat. No. 11,304,770), which is a U.S. national stage application under 35 U.S.C. § 371 (c) of International Application No. PCT/US2017/021284, filed Mar. 8, 2017, which application claims priority to and the benefit of the filing date of U.S. Provisional Patent Application 62/305,867, filed Mar. 9, 2016, each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) Aspects of the present disclosure relate to a surgical instrument force transmission mechanism, configured to impart rotation and translational movement to an actuation mechanism coupled to an end effector of the surgical instrument push/pull rod.

INTRODUCTION

(2) Benefits of minimally invasive surgery are well known, and they include less patient trauma, less blood loss, and faster recovery times when compared to traditional, open incision surgery. In addition, the use of teleoperated, computer-assisted surgical systems (e.g., robotic systems that provide telepresence), such as the da Vinci® Surgical System commercialized by Intuitive Surgical, Inc. of Sunnyvale, Calif., is known. Such teleoperated surgical systems allow a surgeon to operate with intuitive control and increased precision when compared to manual minimally invasive surgeries.

(3) Teleoperated surgical systems include one or more surgical instruments or tools. To perform actions directed by a surgeon, the teleoperated surgical system uses connections that permit motion of a surgical instrument, or a component on which a surgical instrument is mounted, in more than one direction. In other words, the connection may be used to provide more than one degree of freedom for the motion of a surgical instrument. Further, the connection may be used to translate motive force from an actuator to the medical instrument or to a component to which the instrument is mounted. Thus, a connection may be required to provide different functions and movements, even if these functions and movements may otherwise conflict with one another from a mechanical or structural sense.

SUMMARY

(4) Exemplary embodiments of the present disclosure solve one or more of the above-mentioned problems and/or demonstrate one or more of the above-mentioned desirable features. Other features and/or advantages may become apparent from the description that follows.

(5) In accordance with at least one exemplary embodiment, a force transmission mechanism for a teleoperated surgical instrument includes a drive pulley, a drive cable operably coupled with the drive pulley, a driven pulley operably coupled with the drive cable, and an actuation member

operably coupled to the driven pulley. The actuation member is configured to transmit force to actuate an end effector of the surgical instrument. Rotational motion of the driven pulley causes translational movement of the actuation element to actuate the end effector.

(6) In accordance with at least one exemplary embodiment, a surgical instrument for a teleoperated surgical system includes a shaft, an end effector disposed at a distal portion of the shaft, and a force transmission mechanism disposed at a proximal portion of the shaft. The force transmission mechanism includes a drive pulley, a drive cable operably coupled with the drive pulley, a driven pulley operably coupled with the drive cable, and an actuation member operably coupled to the driven pulley. The actuation member is configured to transmit force to actuate an end effector of the surgical instrument. Rotational motion of the driven pulley causes translational movement of the actuation element to actuate the end effector.

(7) In accordance with at least one exemplary embodiment, a method of operating a surgical instrument includes winding a portion of a drive cable over a drive pulley by rotating the drive pulley responsive to a torque applied to the drive pulley, rotating a driven pulley by unwinding another portion of the drive cable from the driven pulley responsive to winding the portion of the drive cable over the drive pulley, translating an actuation element responsive to rotating the driven pulley, and operating an end effector of the surgical instrument responsive to translating the actuation element.

(8) Additional objects, features, and/or advantages will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the present disclosure and/or claims. At least some of these objects and advantages may be realized and attained by the elements and combinations particularly pointed out in the appended claims.

(9) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the claims. The claims should be entitled to their full breadth of scope, including equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate one or more exemplary embodiments of the present teachings and together with the description serve to explain certain principles and operation. In the drawings,

(2) FIG. 1 is a front view of an exemplary embodiment of a patient side cart of a teleoperated surgical system;

(3) FIG. 2 is a perspective view of an exemplary embodiment of a surgical instrument including a force transmission mechanism;

(4) FIG. 3 is a schematic perspective view of an exemplary embodiment of a force transmission mechanism coupled with an actuation member (shown in partial view);

(5) FIG. 4 is a perspective view of an exemplary embodiment of a force transmission mechanism and portion of a surgical instrument shaft;

(6) FIG. 5 is a perspective view of various mechanical components of the force transmission mechanism of FIG. 4 shown in isolation and coupled with an actuation member (shown in partial view); and

(7) FIG. 6 is a perspective view of an exemplary embodiment of a surgical instrument end effector coupled with an actuation member.

DETAILED DESCRIPTION

(8) Various exemplary embodiments herein may be implemented using a da Vinci® Surgical

System (specifically, a Model IS4000, marketed as the da Vinci Xi® Surgical System), commercialized by Intuitive Surgical, Inc. of Sunnyvale, Calif. Persons of ordinary skill in the art will understand, however, that inventive aspects disclosed herein may be embodied and implemented in various ways, including computer-assisted teleoperated and manual embodiments and implementations. Implementations on da Vinci® Surgical Systems are merely exemplary and are not to be considered as limiting the scope of the inventive aspects disclosed herein.

(9) Various exemplary embodiments of the present disclosure contemplate a remotely-controllable surgical instrument having a force transmission mechanism configured to convert input rotary motion to a translational movement of a push/pull rod, cable, or other actuation element.

Exemplary embodiments of the present disclosure also contemplate such a force transmission mechanism for a teleoperated surgical instrument.

(10) FIG. 1 is a front view of the patient side cart **100** of a teleoperated surgical system. A teleoperated surgical system allows diagnostic and corrective surgical procedures to be performed on a patient. Such a teleoperated surgical system is described in U.S. Pat. No. 8,545,515 (filed Nov. 13, 2009), which is hereby incorporated by reference in its entirety. The patient side cart includes a base **102** that rests on the floor, a support tower **104** that is mounted on the base **102**, and several arms that support surgical tools (which can include a stereoscopic endoscope). According to an exemplary embodiment, surgical tools may be arranged according to the embodiments described in U.S. Pat. No. 6,394,998 (filed Sep. 17, 1999) and U.S. Pat. No. 6,817,974 (filed Jun. 28, 2002), which are hereby incorporated by reference in their entirety.

(11) As shown in FIG. 1, arms **106a**, **106b** are instrument arms that support and move the surgical instruments used to manipulate tissue, and arm **108** is a camera arm that supports and moves the endoscope. FIG. 1 also shows an optional third instrument arm **106c**. FIG. 1 further shows interchangeable surgical instruments **110a**, **110b**, **110c** mounted on the instrument arms **106a**, **106b**, **106c**, and an endoscope **112** mounted on the camera arm **108**, which may be interchangeable with a surgical instrument. A surgical instrument **110a** may be mounted to an arm **106a** via a patient-side manipulator (“PSM”) portion **120** that supports and moves the surgical instrument.

(12) FIG. 2 is a perspective view of an exemplary embodiment of a surgical instrument **200**. As shown, the surgical instrument **200** includes a force transmission mechanism **210**, an end effector **220** at a distal end **224** of the surgical instrument **200**, and a shaft **222** connecting the force transmission mechanism **210** and the end effector **220**. The surgical instrument **200** includes one or more members to translate force between the force transmission mechanism **210** and the end effector **220**. For instance, one or more actuation member(s) **226** may connect the force transmission mechanism **210** to the end effector **220** to provide actuation forces to the end effector **220**, such as by extending through an interior of the shaft **222**. By utilizing actuation member(s) **226**, the force transmission mechanism **210** actuates the end effector **220** to, for example, control a jaw of the end effector **220** (or other moveable part of the surgical instrument). Further, because the end effector **220** may be coupled to the shaft **222**, force translated from the force transmission mechanism **210** to the end effector **220** may in turn be translated to the shaft **222**, such as when force transmission mechanism **210** actuates end effector **220** in a rolling motion by rolling the shaft.

(13) Actuation member(s) **226** may be in the form of tension elements, such as when the force transmission mechanism **210** is a pull-pull mechanism, or in the form of one or more force isolation rods, such as when force transmission mechanism **210** is a push-pull mechanism, such as a drive rod element, as described in U.S. Pat. No. 8,545,515, referenced above.

(14) The force transmission mechanism **210** may include one or more components to engage with a patient side cart **100** to translate a force provided by the patient side cart **100** to the surgical instrument **200**. According to an exemplary embodiment, the force transmission mechanism **210** may include one or more interface disks **212**, **214** that engage with the PSM **120** of a patient side cart **100**. Thus, interface disks **212**, **214** may couple with drive mechanisms (e.g.,

servomechanisms) (not shown) in the PSM **120** and translate a force from the drive mechanisms (e.g., servomechanisms) to the surgical instrument **200**. Thus, the interface disks **212**, **214** utilize the actuation forces from the PSM **120** to actuate the instrument **200** through the force transmission mechanism **210** and actuation member(s) **226**. For instance, in an exemplary embodiment, the first disk **212** may be configured to provide a rolling motion to the shaft **222** and provide a roll degree of freedom (“DOF”) for the end effector **220**, while the second disk **214** may operate other DOFs of the end effector **220**, such as, for example, to open and close a jaw mechanism of the end effector.

(15) Force transmission mechanisms, such as the force transmission mechanism **210** of FIG. 2, may be required to transmit relatively large forces between the PSM **120** and the end effector **220** or other movable component of the surgical instrument. For example, the end effector **220** may be in the form of opposing openable and closable jaws such as are used in forceps, grippers, and clamps for closing ligation clips (i.e., a clip applier), etc. Further, actuation of the end effector **220** may require a linear, translational movement of the member **226**, e.g., when the member **226** is a drive rod, while the movement of the interface disks **212**, **214** may be rotational movement. For these reasons, the force transmission mechanism **210** may be required to convert the input rotational movement of at least one of the interface disks **212**, **214** to generally linear, translational movement of a drive rod to actuate the end effector **220**. Thus, it is desirable to provide a force transmission mechanism capable of efficiently transmitting relatively large forces between the interface disks **212**, **214** and the end effector **220**, while at the same time converting rotary motion of an interface disk to translational motion of a drive rod.

(16) Referring now to FIG. 6, an exemplary embodiment of an end effector **620** is shown. The end effector **620** may be or include any of the configurations discussed above, such as a clip applier, forceps, grippers, etc. The end effector **620** includes first and second jaws **690** and **691**, respectively. The first and second jaws **690** and **691** are pivotally connected to a clevis **692** at a pivot, such as pinned joint **693**. A distal end of a drive rod **684** includes, or is attached to, a pin **694** oriented orthogonally with respect to the drive rod **684**. Stated another way, the drive rod **684** and pin **694** provide a T-shaped configuration at the distal end portion of the drive rod **684**. Movement (e.g., translation) of the drive rod **684** in the proximal direction (e.g., actuated by force actuation mechanisms **210**, **310**, or **410**, discussed in connection with FIGS. 2, 3, and 4 respectively) causes the pin **694** to travel within cam slots **695** formed in cam extensions **696**, **697**, respectively, of each of the jaws **690** and **691**. Travel of the pin **694** in the proximal direction through the cam slots **695** causes the pin **694** to bear against the cam extensions **696**, **697** within the slots **695**, thereby move the jaws **690**, **691** toward each other into a closed position. Movement (e.g., translation) of the drive rod **684** in the distal direction causes the pin **694** to bear against the cam extensions **696**, **697** within the slots **695** to move the jaws **690**, **691** away from each other and into an open position.

(17) Accordingly, exemplary embodiments of force transmission mechanisms of the present disclosure may be configured to transmit force between an interface disk (e.g., interface disks **212**, **214** of PSM **120**) and an end effector **220** or other distal movable component of a surgical instrument. In some exemplary embodiments, the force transmission mechanisms according to the present disclosure also convert rotational motion of the interface disk to translational movement of an actuation member (e.g., a drive rod).

(18) Referring now to FIG. 3, a simplified, schematic diagram of a force transmission mechanism **310** illustrates exemplary components and mechanical operation of an exemplary embodiment of a force transmission mechanism. Other exemplary embodiments of force transmission mechanisms, such as the force transmission mechanism **410** described below in connection with FIGS. 4 and 5, may have similar mechanical functionality to the force transmission mechanism **310**, as well as additional features and functionality. In the exemplary embodiment of FIG. 3, the force transmission mechanism **310** includes a drive pulley **330**, which may form a portion of a drive pulley assembly **458** as shown in FIGS. 4 and 5, a driven pulley **332**, and a drive cable **334** at least

partially around each of the drive pulley **330** and the driven pulley **332**. In some embodiments, the drive pulley **330** may be configured as a capstan. Rotational axes **336** and **338** of the drive pulley **330** and the driven pulley **332**, respectively, may be oriented in a parallel or non-parallel relationship. In the exemplary embodiment of FIG. **3**, the rotational axis **338** of the driven pulley **332** is oriented approximately perpendicular to the rotational axis **336** of the drive pulley **330** to impart movement to an actuation element rod **348** along a desired direction, as discussed below. In particular, in the orientation of FIG. **3** the drive pulley **330** has a vertical axis of rotation and the driven pulley **332** has a horizontal axis of rotation.

(19) The drive cable **334** may be a single cable wrapped at least partially around the drive pulley **330** and then connected at ends to the driven pulley **332**. Alternatively, the drive cable **334** may comprise two separate cables, each fixed at one end to the drive pulley **330** and at opposite ends to the driven pulley **332**. Providing a drive cable **334** that has two separate cables may facilitate pre-tensioning of the drive cable **334**, as described in more detail below. The drive cable **334** is wrapped around the drive pulley **330** such that a first portion **340** extends from the drive pulley **330** towards the driven pulley **332** and a second portion **342** extends from the drive pulley **330** also towards the driven pulley **332**, but from a position on the drive pulley about diametrically opposite the first portion **340**. As shown in FIG. **3**, the drive cable **334** wraps multiple times around the drive pulley **330** such that the portions **340** and **342** extend from the drive pulley **330** at positions along the drive pulley that are separated by an axial distance. Respective ends of the first portion **340** and second portion **342** of the drive cable **334** are fixed to the driven pulley **332**, and each portion **340** and **342** of the drive cable **334** wraps around at least a portion of the driven pulley **332**. The drive cable **334** may be or include a single filament of material, or multiple filaments braided or twisted together. The drive cable **334** may comprise metal material such as stainless steel, a titanium alloy, tungsten, or other metals, or may comprise a polymer material or any other material having sufficient tensile strength and flexibility.

(20) Rotation of the drive pulley **330** in direction **344** about the rotational axis **336** causes the first portion **340** to wind onto the drive pulley **330** and the second portion **342** to pay out (unwind) from the drive pulley **330**. Tension generated in the first portion **340** imparts rotation of the driven pulley **332** about the driven pulley rotational axis **338** in direction **346**. As this occurs, the second portion **342** pays out from the drive pulley **330** and winds onto the driven pulley **332**.

(21) Conversely, when the drive pulley **330** is rotated opposite the direction **344** about the rotational axis **336**, the second portion **342** of the drive cable **334** is wound onto the drive pulley **330**, and the first portion **340** pays out from the drive pulley **330**. Tension in the second portion **342** of the drive cable **334** imparts rotation of the driven pulley **332** in a direction opposite direction **346**, causing the first portion **340** to be wound onto the driven pulley **332** while the second portion **342** pays out from the driven pulley **332**. In this manner, rotational motion of the drive pulley **330** is transmitted via the drive cable **334** to the driven pulley **332**. Depending on the relative sizes of the drive pulley **330** and driven pulley **332**, and how the drive cable **334** is wrapped around each, a torque conversion may occur from the rotary input at the drive pulley **330** to the rotary output at the driven pulley **332**.

(22) The driven pulley **332** may be operably coupled to a proximal end **350** of an actuation element, such as a push-pull rod **348**, a distal end (not shown in FIG. **3**) of which is operably coupled with and configured to actuate a distal component of the surgical instrument, such as an end effector, e.g., end effector **220** (FIG. **2**) or end effector **620** (FIG. **6**). As shown in FIG. **3**, the actuation rod **348** can be coupled to the driven pulley **332** with, e.g., a joint such as a ball joint (not shown). Rotation of the driven pulley **332** about the rotational axis **338** causes translational movement of the rod **348** along longitudinal axis **354** along a proximal-distal direction, as the rotating driven pulley **332** applies a force to the rod **348** through the ball joint **352**. Translation of the actuation rod **348** actuates the end effector **220** (FIG. **2**) or **620** (FIG. **6**), as discussed herein in connection with FIGS. **2** and **6**.

(23) FIG. 4 shows an exemplary embodiment of a portion of a force transmission mechanism **410**. The embodiment shown in FIG. 4 may have functionality similar to that previously described and shown in FIG. 3, with specific features and aspects in addition to those discussed in connection with FIG. 3. For example, the force transmission mechanism **410** may include a chassis **456** to which various components of the force transmission mechanism **410** are attached. While the chassis **456** is shown in FIG. 4 with mechanical components of the force transmission mechanism **410** exposed for clarity, the chassis **456** may further include a cover (not shown) configured to engage with the depicted chassis to enclose and protect the mechanical components of the force transmission mechanism **410** from foreign objects, contamination, etc. during use. The chassis **456** may be configured to be attached to a PSM **120** (FIG. 1), and may include interface disks (not shown) such as interface disks **212**, **214** (FIG. 2) configured to interact with mechanisms of the PSM **120**.

(24) An instrument shaft **422** is coupled to the chassis **456** at a proximal end of the instrument shaft **422**. A distal end (not shown) of the instrument shaft **422** includes an end effector, such as end effector **220** shown in FIG. 2. The instrument shaft **422** may be configured to be rotatable relative to the housing portion **456**, e.g., based on a rotational input to an interface disk, such as one of the interface disks **212**, **214** (FIG. 2). An actuation element, such as push-pull rod **348** (FIG. 3), extends at least partially through the instrument shaft **422** and is configured to actuate the end effector positioned at the distal end of instrument shaft **422**, or another component of the instrument shaft **422**. The rod may actuate the end effector as it translates in the proximal-distal direction described above in connection with FIGS. 2 and 6.

(25) The force transmission mechanism **410** includes a drive pulley assembly **458** comprising a drive pulley (capstan) **430** rotationally coupled with the chassis **456**. A first drive cable **440** and a second drive cable **442** are fixed at respective first ends to the drive pulley **430**. The first and second drive cables **440**, **442** are fixed at respective second ends to the drive pulley **430** and fixed at respective second ends to a driven pulley **432**. Rotation of the drive pulley **430** results in corresponding rotation of the driven pulley **432** in a manner similar to that described above in connection with FIG. 3. Cable guides (e.g., fenders) **443** ensure that the first and second actuator cables **440**, **442** do not derail from idler pulleys **560**, which are discussed further below in connection with FIG. 5. The drive rod **348** is coupled with the driven pulley **432** with a ball joint (not shown), discussed below in connection with FIG. 5. A retainer element **457** may retain the actuation member drive rod **348** of FIG. 3 to the driven pulley **432**.

(26) Referring now to FIG. 5, mechanical components of the force transmission mechanism **410** are shown without the housing **456** shown in FIG. 4 to simplify illustration of the various mechanical components. The drive pulley assembly **458** may include or be coupled with a drive input disk **559**, which may be meshed with a drive output disk (not shown) or otherwise rotationally coupled with one or more interface disks (such as interface disks **212**, **214** shown in FIG. 2) actuated by the PSM **120** (FIG. 1) to rotate the drive pulley assembly **458**.

(27) The force transmission mechanism may include components configured to route the first and second actuator cables **440**, **442** between the drive pulley **430** and the driven pulley **432** and ensure that the actuator cables **440**, **442** wind on and off the drive pulley **430** and driven pulley **432** correctly. For example, the force transmission mechanism may include idler pulleys **560** about which the first and second actuator cables **440**, **442** are directed. As shown, for example, cable **442** extends over one idler pulley **560** and cable **440** extends under the other idler pulley **560**. The idler pulleys **560** may be positioned to ensure the first and second actuator cables **440**, **442** extend from the drive pulley **430** at angles substantially perpendicular to the rotational axis **536** of the drive pulley **430**. Similarly, the idler pulleys **560** may be positioned to ensure the first and second actuator cables **440**, **442** extend from the driven pulley **432** at angles substantially perpendicular to the rotational axis **538** of the driven pulley **432**. Such an arrangement may increase (e.g., maximize) the force transmission capability of the force transmission mechanism **410** by providing

an optimal geometric relationship between the actuator cables **440**, **442** and the drive and driven pulleys **430**, **432**. Such a geometric relationship may reduce (e.g., minimize) loss. While the exemplary embodiment of FIG. 5 includes idler pulleys **560**, other components such as guides, pins, or any other routing device may be used to align the actuator cables **440**, **442** with the drive pulley **430** and driven pulley **432** to optimize force transmission between the drive and driven pulleys **430**, **432**.

(28) The drive pulley **430** and the driven pulley **432** may include grooves **562**, **563** in which the actuator cables **440**, **442** are seated to ensure the actuator cables **440**, **442** remain routed correctly around the respective outer diameters of the drive pulley **430** and driven pulley **432**. The driven pulley **432** may be rotationally coupled with the chassis **456** (FIG. 4) with a pin **564**. The pin **564** may be supported directly by the chassis **456**, or may rotate on plain bearings, ball bearings, roller bearings, etc. disposed within the housing **456**. The drive pulley assembly **458** may be mounted within bearings **565** disposed within the housing **456** (FIG. 4).

(29) The first and second actuator cables **440**, **442** may be fixed to the driven pulley **432** by first crimping, soldering, or otherwise affixing enlarged ends (only enlarged end **566** of the first actuator cable **440** shown in FIG. 5 due to perspective of the drawing) to ends of the first and second actuator cables **440**, **442**, and pressing the respective enlarged ends into recesses **568** within the driven pulley **432**. The first and second actuator cables **440**, **442** are partially wrapped around the grooves **562** of the driven pulley **432** and routed over the idler pulleys **560** as described above.

(30) According to an exemplary embodiment, the drive pulley assembly **458** may include features configured to enable pre-tensioning of the first and second actuator cables **440**, **442** during assembly of the force transmission mechanism. For example, in the exemplary embodiment of FIG. 5, the drive pulley **430** includes a lower sheave **570** and an upper sheave **572**. The lower sheave **570** and upper sheave **572** are configured to rotate independently during a pre-tensioning operation (e.g., during assembly of the force transmission mechanism **410**), and to rotate together during operation of the force transmission mechanism. To achieve this, the upper sheave **572** includes a split collar **574** with a clamp screw **576** that can be tightened to rotationally lock the upper sheave **572** with the lower sheave **570**. The first actuator cable **440** may include an enlarged end, such as barrel end **578**, that can be crimped, soldered, welded, or otherwise fastened to an end of the first actuator cable **440**. The barrel end **578** may be inserted within a recess **579** of the lower sheave **570**, such that rotation of the lower sheave **570** winds a portion of the first actuator cable **440** around the lower sheave **570**. The driven pulley **432** may be held in place while a torque is applied to the lower sheave **570** to generate a tension in the first actuator cable **440**. A similar enlarged end of the second actuator cable **442** (not shown due to the perspective of FIG. 5) may be inserted within a similar recess (also not shown) of the upper sheave **572**, and the second actuator cable **442** can be wound and pre-tensioned by rotating the upper sheave **572** in the opposite rotational direction relative to the pre-tensioning torque applied to the lower sheave **570**. The clamp screw **576** is then tightened, rotationally locking the lower sheave **570** and the upper sheave **572** together and maintaining the tension in both the first and second actuator cables **440**, **442**.

(31) In an exemplary embodiment, the driven pulley **432** includes a socket **580** configured to accept a ball joint **582**. The ball joint **582** connects to the actuation (drive) rod **584** operably connected to an end effector (e.g., end effector **220** shown in FIG. 2 or end effector **620** shown in FIG. 6). The ball joint **582** and the drive rod **584** may be free to rotate about a longitudinal axis of the rod **584** within the ball socket **580**. For example, the instrument shaft **422** (FIG. 4) may be configured to rotate along a longitudinal axis thereof in response to an input from one or more interface disks, such as interface disks **212**, **214** (FIG. 2). An end effector (e.g., end effector **220** or end effector **620**) may be coupled to the end of the instrument shaft **422** and operably coupled with the drive rod **584**, e.g., such that translation of the drive rod **584** actuates the end effector as described above. The drive rod **584**, the instrument shaft **422**, and the end effector may rotate in unison based on an input from an interface disk. As the instrument shaft **422**, end effector, and drive rod **584** rotate, the

ball joint **582** rotates within the ball socket **580**. Such a ball joint arrangement, and methods and devices for fastening the ball joint to the drive rod, are described in U.S. Patent App. Pub. No. US 2014/0338477 A1 (filed May 13, 2014), the entire disclosure of which is incorporated by reference herein.

(32) The drive pulley **430** and the driven pulley **432** may be configured to provide a mechanical advantage between the input to the drive pulley assembly **458** (e.g., an input torque applied by a disk interface to the drive pulley assembly **458** to actuate the force transmission mechanism) and the output to the end effector (e.g., end effector **220** (FIG. 2) or **620** (FIG. 6)). For example, a diameter of the drive pulley **430**, a diameter of the driven pulley **432**, and a distance of the ball joint **582** from the rotational axis of the driven pulley **432**, may together define a mechanical advantage to deliver a force sufficient to actuate the end effector. For example, in an embodiment in which the end effector is a jaw mechanism configured to apply ligation clips, e.g., to blood vessels of a patient, (e.g., the end effector comprises a “clip applier”), the end effector may require a force of up to approximately 45 pounds (45 lbf; 200 N) to be delivered by the actuation rod **584**, for example to close the jaws as described above with reference to the exemplary embodiment of FIG. 6. The drive pulley **430** and the driven pulley **432** may have different diameters to provide a desired mechanical advantage between an input to the drive pulley assembly **458** and an output of the driven pulley **432**. For example, a tensile force applied to the first or second drive cables **440**, **442** by rotation of the drive pulley assembly **458** may deliver a force at the actuation rod **584** equal to the tensile force applied to the cables multiplied by a factor greater than one. In an exemplary embodiment, the drive pulley assembly **458** is configured to apply a nominal tensile force of about 30 lbf to the first or second actuator cable **440**, **442** as each cable is wound around the drive pulley assembly **458**, and the mechanical advantage factor may be equal to about 1.5, thereby delivering about 45 lbf/200 N to the actuation rod **584** to actuate the end effector. Other forces and factors of mechanical advantage as required by the specific application, type of end effector, etc. are within the scope of the disclosure, and the mechanical advantage may be tailored to the specific application by varying the relative size of the diameters of the drive pulley **430** and driven pulley **432** and the distance from the rotational axis of the driven pulley **432** to the ball joint **582**.

(33) In addition, the mechanical advantage provided by the force transmission mechanism **410** may vary depending on the rotational position of the driven pulley **432**. For example, the mechanical advantage may be greatest when the ball joint **582** is horizontally aligned (in the orientation view of FIG. 5) with the rotational axis **538** of the driven pulley **432**, as the length of an effective lever arm **586** acting on the ball joint **582** is at a maximum. Stated another way, the lever arm **586** is at a maximum effective length when the lever arm **586** is orthogonal to both the rotational axis **538** of the driven pulley **432** and the longitudinal axis of the drive rod **584**. As rotation of the driven pulley **432** moves the ball joint **582** away from the position of maximum effective length, a maximum force delivered to the end effector is reduced.

(34) Accordingly, in exemplary embodiments, the rotational position of the driven pulley **432** and the ball joint **582** may be optimized such that the mechanical advantage delivered by the force transmission mechanism **410** is at a maximum when the end effector or other actuated component of the surgical instrument is in a position requiring the maximum applied force. For example, in embodiments in which the end effector comprises a jaw mechanism configured as a clip applier (see, e.g., FIG. 6), the required force may be greatest when the jaws of the end effector approach a closed position. Consequently, the force transmission mechanism may be configured such the lever arm **586** approaches the maximum effective length when the jaws of the end effector approach a closed position. In other exemplary embodiments, such as embodiments in which the end effector is configured as forceps, grippers, or other tools, the position in which the lever arm **586** is at the maximum effective length may or may not correspond to a closed position of the jaws of the end effector. For example, in an exemplary embodiment in which the end effector **220**, **620** comprises a dissecting instrument, the rotational position of the driven pulley **432** and the ball joint **582** may be

selected such that the mechanical advantage delivered by the force transmission mechanism **410** provides maximum force to open the jaws from a closed position. As an additional non-limiting example, with some end effector tools, such as bipolar cautery forceps, a possibility exists that tissue may stick to grippers of the tool, and thus require relatively high forces to open the tool. In such applications, the mechanical advantage may be maximized for the force available to open the tool over the force to close the tool, and the drive rod **584** and ball joint **582** may be configured accordingly. Likewise, those of ordinary skill in the art would appreciate that force transmission mechanisms may be configured to transmit maximum force over other ranges of motions when used to actuate other actuatable surgical instrument components, such as for example, articulatable wrist mechanisms, etc.

(35) In exemplary embodiments, the actuation rod **584** may comprise a resilient material configured to deform elastically in a transverse direction **588** as the driven pulley **432** rotates. For example, in addition to translational movement along the longitudinal axis of the actuation rod **584**, the ball joint **582** and the rod **584** near the ball joint **582** may be displaced along an arc as the driven pulley **432** rotates, the ball joint **582** being constrained by ball socket **580** to a circular movement. Stated another way, as the driven pulley **432** rotates clockwise as viewed in FIG. 5, the ball joint **582** may move downward and slightly toward the drive pulley assembly **458**. However, the radius of the arc may be relatively large compared to the translational distance of the ball joint and rod along the longitudinal axis of the rod **584**, and some elastic deformation of the rod **584** may permit movement of the ball joint **582** along the arc as the rod **584** translates. Suitable materials for the drive rod **584** may include relatively elastic materials such as stainless steel, titanium, nitinol, or other metal alloys, polymer materials, or other materials.

(36) The force transmission mechanisms disclosed herein may be desirable over other configurations. For example, compared to various toothed gear arrangements (e.g., helical gear and rack assembly), exemplary embodiments of the disclosure exhibit lower friction and thus higher force transmission efficiency. For example, in some situations intermeshing toothed gear arrangements may exhibit 50 percent or less force transmission efficiency, due to frictional losses and losses attributable to non-optimal geometry of the gear mating surfaces. Embodiments of the disclosure can exhibit force transmission efficiencies above 50 percent, above 75 percent, etc. Additionally, embodiments of the disclosure can exhibit a range of motion of the actuation rod **584** greater than a range of motion obtainable with a helical gear and rack configured to fit within a similar enclosure. Finally, under high applied torque, helical gears and components supporting and positioning the helical gears may deflect, allowing the intermeshing teeth of components (and actuation member) to momentarily disengage and “skip” teeth, leading to misalignment of the mechanism, damage to gear teeth and other components, etc. Thus, embodiments of the disclosure may promote the reliability and functionality of the force transmission mechanism, while permitting an overall compact size to be implemented.

(37) This description and the accompanying drawings that illustrate exemplary embodiments should not be taken as limiting. Various mechanical, compositional, structural, electrical, and operational changes may be made without departing from the scope of this description and the invention as claimed, including equivalents. In some instances, well-known structures and techniques have not been shown or described in detail so as not to obscure the disclosure. Like numbers in two or more figures represent the same or similar elements. Furthermore, elements and their associated features that are described in detail with reference to one embodiment may, whenever practical, be included in other embodiments in which they are not specifically shown or described. For example, if an element is described in detail with reference to one embodiment and is not described with reference to a second embodiment, the element may nevertheless be claimed as included in the second embodiment.

(38) For the purposes of this specification and appended claims, unless otherwise indicated, all numbers expressing quantities, percentages, or proportions, and other numerical values used in the

specification and claims, are to be understood as being modified in all instances by the term “about,” to the extent they are not already so modified. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

(39) It is noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the,” and any singular use of any word, include plural referents unless expressly and unequivocally limited to one referent. As used herein, the term “include” and its grammatical variants are intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that can be substituted or added to the listed items.

(40) Further, this description's terminology is not intended to limit the invention. For example, spatially relative terms—such as “beneath,” “below,” “lower,” “above,” “upper,” “proximal,” “distal,” and the like—may be used to describe one element's or feature's relationship to another element or feature as illustrated in the figures. These spatially relative terms are intended to encompass different positions (i.e., locations) and orientations (i.e., rotational placements) of a device in use or operation in addition to the position and orientation shown in the figures. For example, if a device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be “above” or “over” the other elements or features. Thus, the exemplary term “below” can encompass both positions and orientations of above and below. A device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(41) Further modifications and alternative embodiments will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the systems and the methods may include additional components or steps that were omitted from the diagrams and description for clarity of operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to be understood that the various embodiments shown and described herein are to be taken as exemplary. Elements and materials, and arrangements of those elements and materials, may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the present teachings may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes may be made in the elements described herein without departing from the spirit and scope of the present teachings and following claims.

(42) It is to be understood that the particular examples and embodiments set forth herein are non-limiting, and modifications to structure, dimensions, materials, and methodologies may be made without departing from the scope of the present teachings.

(43) Other embodiments in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the inventions disclosed herein. It is intended that the specification and examples be considered as exemplary only, with the following claims being provided their full scope of breadth, including equivalents, under the applicable law.

Claims

1. An instrument comprising: an end effector coupled to a shaft; a force transmission mechanism proximal of the end effector; and an actuation mechanism coupled to the force transmission mechanism and the end effector, the actuation mechanism comprising a first actuation member segment and a second actuation member segment comprising a rod; wherein the force transmission mechanism comprises: a housing; a rotatable drive input, a drive pulley coupled to the rotatable

drive input and having a first axis of rotation, and a driven pulley having a second axis of rotation non-parallel to the first axis of rotation of the drive pulley; wherein the first actuation member segment is routed over the drive pulley, wherein the second actuation member segment is operably coupled to the driven pulley, and wherein, in response to rotation of the rotatable drive input: at least a portion of the first actuation member segment is wound onto the first drive pulley and unwound from the driven pulley, and the second actuation member segment is translated relative to the housing to impart movement to the end effector.

2. The instrument of claim 1, wherein the end effector comprises a jaw assembly.

3. The instrument of claim 2, wherein the jaw assembly is configured to move between an open position and a closed position.

4. The instrument of claim 1, wherein the first axis of rotation and the second axis of rotation are orthogonal to one another.

5. The instrument of claim 1, wherein the force transmission mechanism is configured to be coupled to a surgical manipulator.

6. The instrument of claim 5, wherein the rotatable drive input comprises an interface element configured to engage the surgical manipulator to receive a rotational force from the surgical manipulator.

7. The instrument of claim 1, wherein the first actuation member segment comprises a cable.

8. The instrument of claim 1, wherein the rod has a longitudinal axis parallel to a longitudinal axis of the drive pulley.

9. The instrument of claim 1, further comprising an idler pulley, wherein the first actuation member segment is routed over the idler pulley from the drive pulley to the driven pulley.

10. A method of imparting movement to an end effector of an instrument, the method comprising: rotating a rotatable drive input of a force transmission mechanism operably coupled to a first actuation member segment; and in response to rotation of the rotatable drive input: winding the first actuation member segment onto a drive pulley about which the first actuation member segment is routed and unwinding the first actuation member segment from a driven pulley having an axis of rotation non-parallel with an axis of rotation of the drive pulley; and translating a second actuation member segment relative to a housing of the force transmission mechanism to impart movement to the end effector, wherein the second actuation member segment comprises a rod.

11. The method of claim 10, wherein the axis of rotation of the driven pulley is orthogonal to the axis of rotation of the drive pulley.

12. The method of claim 10, wherein the end effector comprises a jaw assembly configured to move between an open position and a closed position.

13. The method of claim 10, further comprising coupling the force transmission mechanism to a manipulator system.

14. The method of claim 13, wherein rotating the rotatable drive input of the force transmission mechanism comprises rotating the rotatable drive input of the force transmission mechanism with the manipulator system.
